

THE SUPPLEMENTAL INFORMATION FOR *JCE ONLINE*

CAS Registry numbers

Acetone	67-64-1
Ammonia	7664-41-7
Ammonium bromide	12124-97-9
Ammonium carbonate	506-87-6
Ammonium chloride	12125-02-9
Barium chloride	10361-37-2
Benzylamine	100-46-9
Boric acid	10043-35-3
Calcium carbonate	471-34-1
Chromium(III) chloride	10025-73-7
Copper(II) acetate	142-71-2
Cobalt(II) carbonate	7542-09-8
Cobalt(II) chloride	7646-79-9
Copper(II) sulfate	7758-98-7
Devadra's alloy	263014-00-2
Dichloromethane	75-09-2
Ethanol	64-17-5
Ether	60-29-7
Ethylenediamine	107-15-3
Glucose	50-99-7
2-Hydroxyethylamine	141-43-5

Hydrobromic acid	10035-10-6
Hydrochloric acid	7647-01-0
Hydrogen peroxide	7722-84-1
Iron(III) sulfate	10028-22-5
Lead(IV) oxide	1309-60-0
Mercury tetrathiocyanatocobaltate(II)	19756-24-2
Methanol	67-56-1
Nickel(II) acetate	373-02-4
Nickel(II) chloride	7718-54-9
Nitric acid	7697-37-2
Oxalic acid	144-62-7
Potassium bromide	7758-02-3
Potassium carbonate	584-08-7
Potassium chloride	7447-40-7
Potassium dichromate	7778-50-9
Potassium oxalate	583-52-8
Potassium permanganate	7722-64-7
Potassium sulfate	7778-80-5
Salicaldehyde	90-02-8
Selenium	7782-49-2
Silver nitrate	7761-88-8
Sodium hydroxide	1310-73-2
Sodium nitrite	7632-00-0

Sulfuric acid 7664-93-9

Zinc 7440-66-6

Preparation of Group A complexes

Bis-[N-benzylsalicylideneiminato]nickel(II) which is typical of all the complexes of Group A, is prepared as follows.

6.11 g (0.05 mol) of salicylidene is treated with 5.36 g (0.05 mol) of benzylamine in 40 mL of ethanol. The solid Schiff base forms immediately and is filtered using a Whatman #1 paper and recrystallised from ethanol. The Schiff base is then reacted with 7.3 g (0.025 mol) of nickel(II) nitrate in 80 mL water. The olive green precipitate of the coordination complex forms immediately. After cooling in an ice-bath, the solid is filtered using a Whatman #1 paper and recrystallised from ethanol. The yield of complex is 5.7 g. All other complexes in this group are prepared in a similar manner, using the appropriate aldehyde and amine and the correct molar ratio of ligand to metal ion.

Analytical procedures to be followed for each Group of complexes

Group A complexes:

Collect the Infra-red spectrum, collect the UV-visible spectrum, determine the magnetic moment by the Gouy technique, determine the nitrogen content by the Kjeldahl method, the carbon content by the Leco carbon measurement, and the metal ion by FAAS.

Group B complexes:

Collect the Infra-red spectrum, collect the UV-visible spectrum, determine the magnetic moment by the Gouy technique, determine the conductance of a 0.001M solution. For complexes 7-10, determine the nitrogen content by the Kjeldahl method, the metal ion by FAAS, and the halide counter anion by conductance titration with standard silver

nitrate. For complexes 11-15, determine the two metal ions by FAAS and the oxalate content by the Leco method. For complexes 16-18, determine the nitrogen content by the Kjeldahl method, the metal ion by FAAS, and the chloride counter anion by conductance titration with standard silver nitrate.

The following are discussions and hints on the dissolution methods to be followed prior to the FAAS determinations, and the Kjeldahl and Leco experiments.

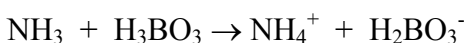
Dissolution of complexes prior to FAAS analysis

All the complexes are digested with concentrated nitric acid prior to FAAS analysis. Consider complex number 9 in Group B as an example. 0.2000 g of complex is digested with 15 mL concentrated nitric acid for 30 min and made up to 1 L in a volumetric flask with deionised water. 10 mL of this solution is then diluted to 100 mL in a volumetric flask with deionised water yielding a 4.5 mg/L^{-1} (ppm) solution ready for FAAS analysis.

Kjeldahl procedure

The standard Kjeldahl procedure involves three fundamental steps:

1. The addition of concentrated sodium hydroxide to liberate ammonia
2. The collection of ammonia in boric acid.



3. Titration against standard hydrochloric acid.



Possible variations relevant to transition metal complexes

1. If there are only NH_3 ligands e.g. $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$, in the complex, simply introduce the sodium hydroxide solution and distil.

2. Complexes containing C-N or C=N bonds, that is the Group A complexes and the ethylenediamine complexes of Group B, must be first broken down by digestion with concentrated sulfuric acid in the presence of potassium sulfate (boiling point elevation) and selenium catalyst to form the ammonium hydrogensulfate salt, prior to undertaking the standard procedure.
3. Complexes containing both NH_3 and $\text{NO}_3^-/\text{NO}_2^-$ ligands, firstly use the standard procedure to analyse the NH_3 ligands. To determine the $\text{NO}_3^-/\text{NO}_2^-$ ligands, Devarda's alloy (45% Al, 5% Zn, 50% Cu) is then added to the same flask, where the alloy in the presence of the caustic reduces the NO_3^- and/or NO_2^- groups to NH_3 . The ammonia is then determined by the standard procedure. It is important to note that the Kjeldahl procedure does not distinguish between the nitrite and nitrate ligand.

Hints for Kjeldahl procedure

Before commencing it is important that the quantity of sample required to give a suitable titration volume be calculated. It also may be more efficient to do two titrations from one digest rather than two individual digestion/titration procedures. For samples requiring digestion in sulfuric acid it is important to cool the solutions and dilute with a small amount of water before adding the NaOH. This can be done for an individual digest/titrate by adding water to the reaction-distillation flask and then adding the NaOH solution. For two titrations from the one digest, dilute accurately to 50 mL, then transfer a 20 mL aliquot to the distillation flask and finally add the NaOH solution. Aim for a 10 to 15 mL final titration. It is also important to note that the larger the sample the more digestion time required and typically the sample should be < 0.5 g of compound. However, for complexes of Group A, where the %N is low, this is not feasible and you

will need to use a larger reaction flask (e.g. 400 mL) for the digestion and heat it more aggressively.

Leco gravimetric analysis

The content of carbon is determined as carbon dioxide. The sample crucible is placed into a quartz combustion tube and heated by an induction furnace under a stream of purified oxygen. At the elevated temperatures the oxygen combines with the carbon in the sample to form carbon dioxide which is collected in an absorption bulb.

Weigh into a Leco crucible the following items according to the carbon content of your compound. Place a Leco cover on top of the crucible before putting it in the analyzer.

Standardisation

Masses of material required for complexes of high carbon content (those of Group A)

Accurately weighed 0.02-0.03 g glucose together with 0.8 g iron, 0.4 g copper and 0.4 g tin accelerators covered with silica.

Masses of material required for complexes of low/medium carbon content (the oxalato and ethylenediamine complexes of Group B)

Accurately weighed 0.08 g calcium carbonate together with 0.8 g iron, 0.4 g copper and 0.4 g tin accelerators covered with silica.

Sample Determination

Repeat the above procedure for the complex, but with the mass of sample determined from the following table.

% Carbon expected	Mass of sample (g)
70	0.015
40	0.025
12	0.08
7	0.15
4	0.25

